

HARDIK BARAIYA

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PROFESSIONAL SUMMARY

Results-driven **Software Developer in Test / SDET** with **7+ years** of experience architecting automated testing solutions across distributed systems, networking protocols, and cloud-native platforms. Specialized in building **agentic, self-healing test frameworks** and integrating **AI/Copilot-assisted test generation** to accelerate delivery while safeguarding reliability. Proven track record validating DNS and TCP/IP protocol stacks, hardening CI/CD quality gates, and sustaining **99.9% uptime** through SRE-based observability. Adept at leading shift-left strategies, mentoring engineers, and optimizing cloud infrastructure cost and performance at scale.

WORK EXPERIENCE

The Canadian Internet Registration Authority (CIRA)

Ottawa, ON

Software Developer in Test

Jun 2024 – Present

- **Led end-to-end validation strategy for the DNS PIVOT project**, migrating DNS Firewall from Akamai to a self-hosted PowerDNS architecture with custom TCP/IP and DNS protocol stacks; refactored the entire regression suite to validate routing policy enforcement, ensuring zero customer-impacting downtime.
- **Designed and executed functional, scale, and interoperability test plans** for Off-Network Protection (ONP) across Windows, MacOS, Android, and iOS; acted as primary technical liaison with third-party vendors to verify cross-platform protocol compliance and feature parity with in-house networking standards.
- Architected an **agentic, self-healing E2E automation framework** for CiraHub using Playwright MCP and TypeScript; the framework autonomously detects UI/API contract changes, re-generates affected selectors and test steps, and achieves **95% test coverage** with near-zero flakiness.
- Integrated **AI/Copilot-assisted test generation** into the development workflow, using Claude and GitHub Copilot to auto-draft Python test cases from feature specs, accelerating test authoring by **40%** and ensuring coverage parity with rapidly evolving API contracts.
- Engineered a distributed **load and performance testing suite** using Locust and Docker to stress-test the DNS Firewall Device Registration API at 2x production traffic; identified and resolved a critical memory leak that would have impacted SLO compliance at scale.
- Developed a data-driven **visual regression suite** uploading custom Parquet files to AWS S3 for dashboard rendering, with automated pixel-by-pixel graph comparisons ensuring **100%** visual consistency across releases.
- Optimized GitLab CI/CD pipelines with parallel test execution, automated quality gates (SAST/DAST), and dedicated QA environments, significantly cutting build-to-test feedback loops.
- Drove **AWS cost optimization** by automating non-production environment shutdowns using tag-based policies across EC2, ECS, and RDS, reducing idle compute usage by **nearly 60%**.
- Managed 24/7 on-call reliability for DNS Firewall products, maintaining **99.9% system uptime** through proactive Splunk monitoring and **SLI/SLO-based alerting** in Datadog.

ZIZO Technologies, Inc

Remote, Canada

Lead Software Development Engineer in Test (SDET)

Aug 2023 – May 2024

- **Directed a shift-left testing strategy** across the engineering org, identifying 30% more defects during development and accelerating release velocity by **20%** through early-stage test design reviews and defect risk analysis.
- Deployed ReportPortal on Docker/AWS EC2, leveraging **AI-driven failure analytics** to eliminate **15 hours/week** of manual failure triage and pattern-based flakiness classification.
- **Architected a greenfield test platform** using CodeceptJS with Playwright engine; established self-healing locator patterns, layered abstraction for cross-browser testing, and AI-assisted test maintenance to reduce script obsolescence.
- Orchestrated **performance testing strategies** using k6, uncovering critical bottlenecks that enabled the system to scale to **2x previous concurrent user loads**.
- Integrated **automated accessibility (a11y) auditing** into CI pipelines, establishing a WCAG compliance baseline that prevented accessibility regressions across all new UI deliverables.
- Mentored a team of engineers on automation best practices; established coding standards that reduced test code review cycles by **50%**.

ZIZO Technologies, Inc

Remote, Canada

Software Development Engineer in Test (SDET)

Aug 2021 – Jul 2023

- Built and maintained comprehensive **API test suites with Python and Postman**, ensuring 100% reliability across internal microservices; authored **data-driven test scripts** covering functional, boundary, and regression scenarios.

- Reduced manual regression effort by **40%** through strategic implementation of modular automated scripts in Azure DevOps CI/CD pipelines.
- Leveraged Datadog for real-time error tracking and proactive monitoring, decreasing **Mean Time to Resolution (MTTR)** for production issues by **35%**.
- Integrated **automated quality gates** into Azure DevOps pipelines, enforcing a green-build culture and preventing defective code from reaching staging environments.
- Authored **technical documentation and Wiki guides** for the automation framework, streamlining onboarding and reducing ramp-up time for new engineers.

TatvaSoft

Gujarat, India

Associate Software Engineer in Test

Jan 2019 – Jul 2021

- Developed **200+ functional test scenarios** in collaboration with UX/UI and Product teams, ensuring high-quality standards for enterprise clients across web and mobile platforms.
- Automated legacy manual processes using Selenium/TestNG with Python, achieving a **50% reduction** in end-to-end testing time for bi-weekly releases.
- Implemented the **Page Object Model (POM)** to increase test maintainability and reduce script failure rates caused by UI changes.
- Led **cross-browser testing initiatives** using BrowserStack across Chrome, Safari, and Firefox, ensuring application reliability for a global user base.
- Managed defect lifecycles in Jira, maintaining a **98% defect-fix verification rate** before production deployments.

TECHNICAL SKILLS

Languages & Scripting: Python, JavaScript, TypeScript, Java, Bash, SQL

Networking & Protocols: TCP/IP, DNS, HTTP/S, Routing Fundamentals; distributed systems protocol testing

Testing & Automation: Playwright (incl. Playwright MCP), Agentic & Self-Healing Test Frameworks, Selenium WebDriver, CodeceptJS, Postman, Pytest, TestNG, k6, Locust

AI-Assisted Development: Copilot-Assisted Test Generation, Agentic Testing Pipelines, LLM-Powered Failure Analysis

DevOps & CI/CD: GitLab CI/CD, Azure DevOps, Jenkins, Git, DevSecOps (SAST/DAST), Automated Quality Gates

Cloud & Infrastructure: AWS (EC2, ECS, RDS, S3), Docker, Kubernetes, Terraform, CloudFormation, Ansible

Observability & Reliability: Datadog, Splunk, AWS CloudWatch, ReportPortal, SRE Principles (SLIs/SLOs), Performance Analytics

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB

EDUCATION

Carleton University

Ottawa, ON

Master of Electrical & Computer Engineering (Software Engineering)

Apr 2023

Gujarat Technological University

Gujarat, India

Bachelor of Electronics and Communications Engineering

Apr 2019

CERTIFICATIONS

Kubernetes: CN120: Kubernetes Application Essentials (Mirantis)

Cloud/DevOps: The Ultimate Docker Course